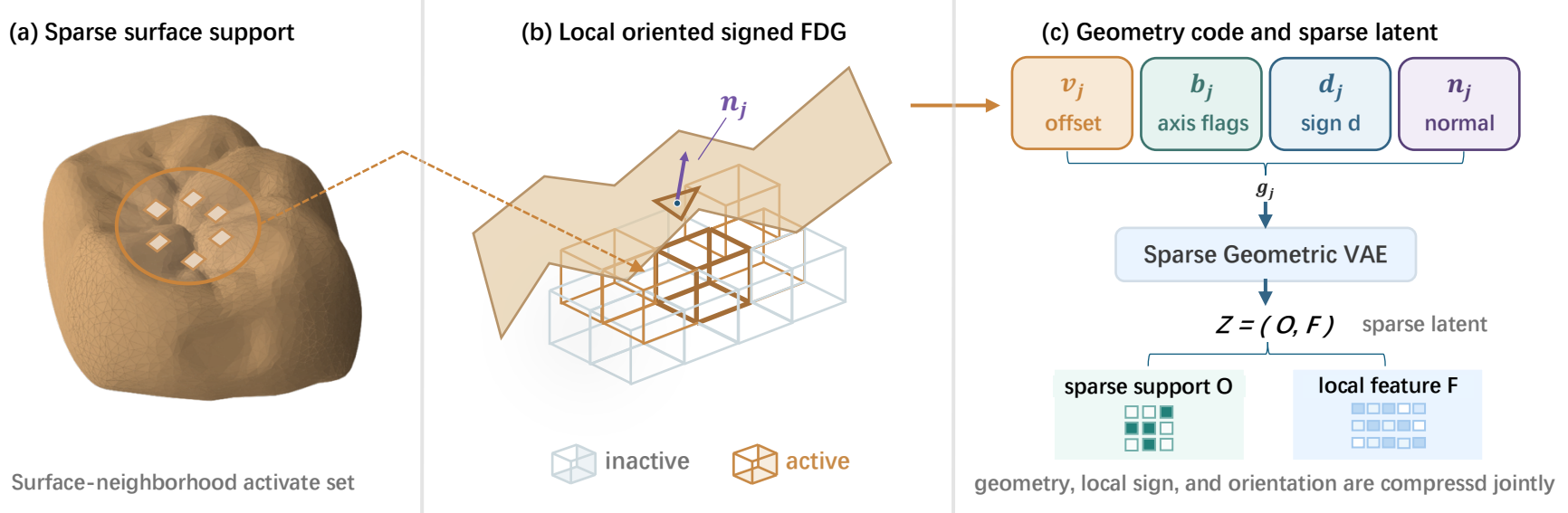

(a) Sparse surface support
(b) Local oriented signed FDG
(c) Geometry code and sparse latent
n_j
v_j offset
b_j axis flags
d_j sign d
n_j normal
g_j
Sparse Geometric VAE
$Z = (O, F)$ sparse latent
sparse support O
local feature F
inactive
active
Surface-neighborhood activate set
geometry, local sign, and orientation are compressd jointly